

Ubuntu - SE4SA531 to SE4SA550 Firmware Upgrade

These instructions start with a fresh Ubuntu installation, install and verify sedutil, perform the PSID revert, and then follow the supplied SE4SA531 to SE4SA550 Linux firmware upgrade procedure.

CAUTION: The PSID revert and firmware upgrade process will erase the drive. Make sure there is no data on the drive that needs to be saved.

1. Install sedutil and NVMe CLI

Open Terminal and install the packages needed to build sedutil, plus NVMe CLI:

```
sudo apt update
sudo apt install -y git build-essential autoconf automake libtool nvme-cli p7zip-full
```

Download and build sedutil:

```
cd ~
git clone https://github.com/Drive-Trust-Alliance/sedutil.git
cd sedutil
autoreconf --install
./configure
make -j"$(nproc)"
```

Install the finished sedutil-cli command so it can be run from any directory:

```
sudo cp sedutil-cli /usr/local/sbin/sedutil-cli
sudo chmod 755 /usr/local/sbin/sedutil-cli
```

2. Verify sedutil and identify the drive

First make sure sedutil runs and can see the installed drives:

```
sudo sedutil-cli --scan
```

Find the NVMe drive that will be upgraded. Use the model and serial number to make sure you have the correct drive.

```
sudo nvme list
lsblk -o NAME,MODEL,SERIAL,SIZE,TYPE
```

Important: The examples below use /dev/nvme0. Replace it with the actual NVMe controller for the drive you are upgrading.

Query the drive with sedutil:

```
sudo sedutil-cli --query /dev/nvme0
```

The query should return the drive's TCG/Opal information without an error. If sedutil cannot query the drive, stop here and resolve that before continuing.

3. Perform the PSID revert

Locate the PSID on the physical drive label. Enter it exactly as printed. For sedutil, the PSID should be entered in all capital letters with no dashes.

Double-check: Confirm the model and serial number one more time before running the next command. This operation erases the target drive.

```
sudo sedutil-cli --yesIreallywanttoERASEALLmydatausingthePSID <PSID> /dev/nvme0
```

A successful PSID revert should report:

```
INFO: revertTper completed successfully.
```

Query the drive again:

```
sudo sedutil-cli --query /dev/nvme0
```

Confirm the drive is back in the inactive state and shows:

```
LockingEnabled = N
```

4. Put the SE4SA550 upgrade file on the Ubuntu Desktop

Copy the supplied firmware archive to the Ubuntu Desktop. The supplied archive is:

```
PCIETool08-5890_DLMC(Seagate)(SE4SA550)_Linux.7z
```

The easiest setup is to create a folder named SE4SA550_Upgrade on the Desktop and keep the firmware files there.

```
mkdir -p ~/Desktop/SE4SA550_Upgrade  
cp "/path/to/PCIETool08-5890_DLMC(Seagate)(SE4SA550)_Linux.7z" ~/Desktop/SE4SA550_Upgrade/  
cd ~/Desktop/SE4SA550_Upgrade
```

If the file is on a USB drive: You can copy it into the SE4SA550_Upgrade folder using the Ubuntu Files application instead of using the cp command. Once copied, continue with the commands below.

Extract the supplied 7-Zip archive:

```
7z x "PCIETool08-5890_DLMC(Seagate)(SE4SA550)_Linux.7z"  
ls
```

After extraction, locate the Linux 64-bit upgrade tool named:

```
PCIETool08-5890_DLMC(Seagate)(SE4SA550)_Linux_64bit
```

If the extracted files were placed inside another folder, cd into that folder before continuing.

5. Make the firmware tool executable

From the directory containing the Linux 64-bit upgrade tool, run:

```
chmod 777 PCIETool08-5890_DLMC\ (Seagate)\ (SE4SA550)\_Linux_64bit
```

6. Run the SE4SA550 firmware upgrade

Run the upgrade tool using the AUTO option:

```
sudo ./PCIETool08-5890_DLMC\ (Seagate)\ (SE4SA550)\_Linux_64bit AUTO
```

Allow the tool to finish. The supplied upgrade procedure says the result should show "Pass". Do not disconnect or power off the drive while the upgrade is running.

7. Verify the new firmware version

After the upgrade reports Pass, check the drive with NVMe CLI:

```
sudo nvme list
```

Find the upgraded drive and confirm that the firmware revision is:

```
SE4SA550
```

If the firmware revision shows SE4SA550, the firmware upgrade is complete.

Quick command sequence

This is the condensed sequence after Ubuntu is installed:

```
sudo apt update
sudo apt install -y git build-essential autoconf automake libtool nvme-cli p7zip-full

cd ~
git clone https://github.com/Drive-Trust-Alliance/sedutil.git
cd sedutil
autoreconf --install
./configure
make -j"$(nproc)"
sudo cp sedutil-cli /usr/local/sbin/sedutil-cli

sudo sedutil-cli --scan
sudo nvme list
sudo sedutil-cli --query /dev/nvme0

sudo sedutil-cli --yesIreallywanttoERASEALLmydatausingthePSID <PSID> /dev/nvme0
sudo sedutil-cli --query /dev/nvme0

mkdir -p ~/Desktop/SE4SA550_Upgrade
cd ~/Desktop/SE4SA550_Upgrade
7z x "PCIET00L08-5890_DLMC(Seagate)(SE4SA550)_Linux.7z"

chmod 777 PCIET00L08-5890_DLMC\Seagate\SE4SA550_Linux_64bit
sudo ./PCIET00L08-5890_DLMC\Seagate\SE4SA550_Linux_64bit AUTO

sudo nvme list
```

Expected end result: the upgrade tool reports Pass and nvme list reports firmware revision SE4SA550.